Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in M>=6.5 Earthquake Catalogs, 1973–2026 CE

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Abstract.

We analyse an analysis catalog of 4,267 unique M>=6.5 events (modern window 1973–2026: 2,041 events; provenance: 4,418 CSV rows, ~151 NOAA M<6.5 excluded from clustering) using the Baiesi–Paczuski metric eta with tectonic-path distance (Bird 2003). The detector yields 47 algorithmic candidates (27 modern); however, ETAS validation (p_ETAS=1.0) shows these candidates are indistinguishable from background activity. Primary result — null result (falsification of the global-series hypothesis): the detector finds on average 27.0 candidates in catalog-calibrated ETAS synthetic catalogs, matching N_obs=27 — not specific to global series; reflects catalog background structure (see Sec. 5.5–5.6). Permutation p=0.0001 (1/10,001) rejects temporal Poisson null only — not teleseismic triggering. Limitations — Sec. 5.5.

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi–Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that M>=7 clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global M>=7 and M>=8 rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. Test (and if warranted, falsify) the hypothesis that physically meaningful multi-regional global series exist, with primary inference on the modern window (1973–2026), using complementary null tests (permutation vs ETAS) and explicit detector liberalness assessment. Scope. We combine nearest-neighbor clustering with heuristic tectonic hint and ETAS validation; this complements global rate tests (Michael 2011; Shearer & Stark 2012) with a different eta-linkage statistic but does not supersede their conclusions.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using +/-30 days and <=50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique M>=6.5 events (from 4,418 CSV rows; ~151 M<6.5 excluded from clustering). USGS ComCat: 2,088 raw (1973–2026) → 2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). quality_score is metadata, not an inclusion filter. Final catalog: 4,267 M>=6.5 events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields M_c = 6.55. Maximum-likelihood b-value from 1,688 events above M_c:

$$b = 0.911 \pm 0.018 \quad (n = 1{,}688 \text{ events})$$

2.2 Heuristic metric with tectonic hint

We tested an alternative to Euclidean distance (Bird 2003 graph). In 98% of pairs the implementation falls back to 1.5x great-circle distance — effectively scaled Euclidean; no improvement for global analysis (failed hypothesis test). r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX). If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists, $r_{ij} = 1.5 \times r_{GC}$ (great-circle Haversine). Fallback audit (analyze_tectonic_fallback.py, fig07): 4987 pairs from 500 events; 98.0% GC fallback, 2.0% Dijkstra (4015 snap>500 km, 872 no path, 100 Dijkstra; 95 materially different; examples FE31 Japan, FE35 Philippines). Metric adds value for ~2% boundary-proximal pairs only. Tectonic diagnostic: median Deltalog10eta = +0.28.

$$\eta_{ij} = t_{ij} \cdot r_{ij}^{1.6} \cdot 10^{(-b \cdot m_i)}$$

b=1.0 per Baiesi & Paczuski (2004) for eta comparability; catalog b=0.911+/-0.018 in Monte Carlo null only.

2.3 Analysis pipeline

Raw catalogs → dedup → exclude M<6.5 (~151 NOAA) → GK declustering (primary) → eta NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: eta NN + KDE threshold). At global M>=6.5 scale ZBZ is permissive (sparse events → high eta). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary, N=27 likely unchanged (untested).

Why GK–ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low eta at global catalog scale. Minimal ZBZ removal confirms declustering is not a critical fork.

2.4 Threshold eta_0, series detection, and ETAS parameters

Automatic eta_0 from nearest-neighbor eta distribution: KDE valley between bimodal $\log_{10}(\text{eta})$ modes (Zaliapin & Ben-Zion 2013); default $\text{eta}_0 = 10^{\text{median log10 eta}}$. Validated against Poisson permutation null (n = 10,000).

1. Declustering via Gardner–Knopoff [1974].
2. Nearest-neighbor forest: parent i* = argmin eta_{ij} subject to $\text{eta}_{ij} < \text{eta}_0$.
3. Global series: N >= 4; M >= 6.5; >= 3 Flinn–Engdahl regions.

4. Sliding windows (1, 2, 5 yr); overlapping groups merged.

ETAS parameters (catalog-calibrated): MLE on 2,041 events (scripts/calibrate_etas.py, results/etas_calibration.json): mu≈0.103, K≈0.495, alpha≈0.063, c≈10^{-4} d, p≈1.36. Literature defaults (Helmstetter & Sornette 2003) retained for comparison.

2.5 Statistical validation

Permutation test: n = 10,000, p <= 0.0001, z = -6.17 (modern). ETAS validation: calibrated params (mu≈0.103, K≈0.495, alpha≈0.063, c≈10^{-4} d, p≈1.36); 1000 catalogs/seed. Single seed=42: FPR=1000/1000; mean 27.0, p_ETAS = 1.0; N_obs=27. Multiseed ETAS (seeds 42–51, n=1000 each, results/etas_multiseed.json): mean=27.0, σ=0.0, FPR=1.0, p_ETAS=1.0 for all 10 seeds — perfect stability because calibrated ETAS matches catalog event rate (~2001 background events). Literature mu=0.008 comparison only (earlier n=100 runs): mean≈15.5, p_ETAS <= 0.001. Detector liberalism — see Sec.5.6 (FPR=1000/1000). Multiple comparisons (Methods): BH post-hoc on N=47, 45/47 at q=0.05 — not discovery. Declustering: GK 2,017/2,041 (primary); ZBZ 2,040/2,041 (sensitivity only).

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 detector candidates (not validated global series): 27 modern (p <= 0.0001), 15 early instrumental (p = 0.007; pre-1960 incompleteness caveat), 5 historical candidates (not significant, p = 0.46; 47 M>=6.5 events pre-1900). 142 cluster candidates before filtering.

ID	N	Reg.	Mmax	Period	lat	lon	Notes
1905–1910	193	43	8.8	1905–1910	-60..72	-180..180	Early instrumental; incomplete before ~1960
S047	53	5	8.0	1982–2024	-21.5..18.9	-175.5..155.2	Western Pacific
S170	46	12	9.1	2002–2023	-14.3..33.8	-113.1..167.3	Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	-8.1..16.5	120.4..146.4	Western Pacific arc
S116	22	5	8.2	1993–2021	-31.7..10.8	-179.5..179.4	South Pacific

Table 1. Top five multi-regional series (>= 3 Flinn–Engdahl regions).

3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median Deltalog10eta = +0.28 on random pairs (mostly 1.5x GC fallback).

4. DISCUSSION AND CONCLUSIONS

The detector finds on average 27.0 candidates in catalog-calibrated ETAS synthetic catalogs, matching N_obs=27 — not specific to global series; reflects catalog background structure. The global-series hypothesis is not supported (see Sec. 5.5). The permutation test rejects only a temporal Poisson null — trivial for earthquake catalogs with aftershocks.

1. Heuristic with tectonic hint: 98% GC fallback — failed hypothesis test.
2. 47 detector candidates indistinguishable from ETAS null; Delta CFS/dynamic stress — future work only.

Future work / supplement: Static Delta CFS and dynamic stress for S170, S047, S095; full ETAS MLE and multi-seed robustness.

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md). External DOI deposition (Zenodo) deferred — GitHub only.

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